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ORIGINAL ARTICLE

Forecasting Methane Emissions in Somalia: A Hybrid Time-Series Approach for Climate and Public Health Insights

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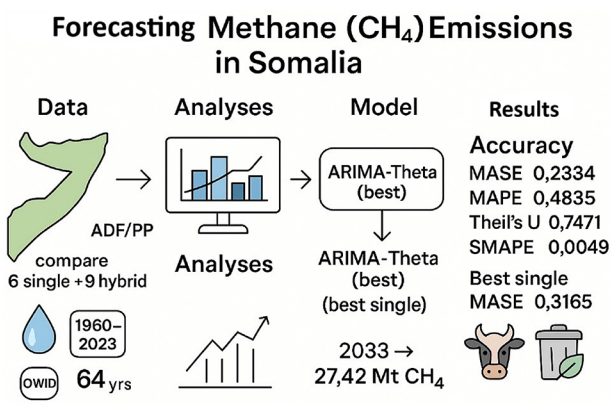
Abstract

This study addresses a critical gap in climate and public health planning for Somalia by identifying the most accurate forecasting model for its annual methane (CH4) emissions. Given Somalia's vulnerability to environmental change and the significant contribution of its agricultural sector (particularly livestock) to CH4 emissions, robust projections are essential for effective mitigation. Utilizing annual time-series data from 1960 to 2023, the research rigorously compared fifteen models, including six individual approaches (ARIMA, ETS, TBATS, Theta, ARFIMA, NNAR) and nine hybrid combinations. Data stationarity was ensured through first-order differencing. Model performance was evaluated using four distinct error metrics: Theil's U, Mean Absolute Percentage Error (MAPE), Symmetric Mean Absolute Percentage Error (SMAPE), and Mean Absolute Scaled Error (MASE). While the ARIMA (0,1,1) model with drift emerged as the best-performing individual model, the hybrid ARIMA-Theta model significantly outperformed all other methods. It demonstrated superior predictive accuracy across all evaluation metrics (Theil's U: 0.7471; MAPE: 0.4835; SMAPE: 0.0049; MASE: 0.2334). Forecasts from this optimal hybrid model project a continued, substantial increase in Somalia's CH4 emissions, anticipated to reach approximately 27.42 million metric tons by 2033. These findings underscore the superior effectiveness of hybrid models for forecasting complex environmental data like CH4 emissions. The projected rise in emissions highlights an urgent need for national intervention, providing crucial, data-driven insights for Somali policymakers to craft targeted mitigation strategies in agriculture and waste management, integrate climate action into public health planning, and fulfill national and international environmental commitments in a climate-affected and vulnerable nation.

Graphical Abstract



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This study pioneers the first comprehensive forecasting of Methane (CH4) emissions in Somalia, a nation particularly vulnerable to climate change impacts, by analyzing 64 years of annual data from 1960 to 2023. As depicted in the graphical abstract, the research initiates with raw data for Somalia (represented by the country's outline and a droplet for historical data from Our World in Data [OWID]) which then undergoes rigorous analyses, including ADF/PP tests and a comparison of 6 single and 9 hybrid time series models. The central finding, highlighted in the "Model" section, demonstrates that the ARIMA-Theta (best) hybrid model consistently outperformed all others in predictive accuracy. The "Results" section quantitatively supports this, showcasing the ARIMA-Theta's superior performance across key metrics such as MASE (0.2334), MAPE (0.4835), Theil's U (0.7471), and SMAPE (0.0049), notably outperforming the best single model MASE (0.3165). This robust model projects a significant increase in Somalia's CH4 emissions to 27.42 million metric tons by 2033 (visually represented with an upward arrow and the projected value). The "Results" section further underscores the practical implications, with icons representing livestock (cow) and waste management (trash bin with leaf) to indicate the primary sources of these emissions. This research provides crucial data-driven insights for Somali policymakers, enabling the development of targeted mitigation strategies in agriculture and waste management to enhance climate resilience and public health.

Highlights

- First comprehensive forecast of Somalia's CH4 emissions using 15 single and hybrid time series models.
- A hybrid ARIMA-Theta model demonstrates superior forecasting accuracy, outperforming all other tested models.
- Somalia's methane emissions are projected to rise to approximately 27.42 million metric tons by 2033.
- Provides a crucial evidence-base for developing targeted climate and public health policies in Somalia, supporting national climate action.
- Offers data-driven insights for policymakers to craft targeted mitigation strategies in agriculture and waste management.

Keywords Methane (CH4) · Forecasting · Hybrid models · Climate change · Public health · Somali

Abbreviations

ADF	Augmented Dickey-Fuller
AIC	Akaike Information Criterion
ARFIMA	Autoregressive Fractionally Integrated Moving Average
ARIMA	Autoregressive Integrated Moving Average
ETS	Error Trend Seasonality
GHG	Greenhouse Gas
MAPE	Mean Absolute Percentage Error
MASE	Mean Absolute Scaled Error
NNAR	Neural Network Autoregression
PP	Phillips-Perron
SMAPE	Symmetric Mean Absolute Percentage Error
TBATS	Trigonometric Seasonal Decomposition of Time Series (Trigonometric, Box-Cox, ARMA errors, Trend, and Seasonality)
CH4	Methane

1 Introduction

Methane (CH4), a potent greenhouse gas, has seen its atmospheric concentrations more than triple since the industrial revolution began, significantly contributing to current global warming (Rocher-Ros et al. 2023). Recent studies dealing with atmospheric modeling emphasize the necessity of establishing accurate background levels for such greenhouse gases to better understand their regional variability and contribution to the global carbon budget (Malacaria et al. 2025). Agriculture is the largest anthropogenic source of CH4, releasing about 267 Tg CH4 y^{-1} into the atmosphere in 2024 (Lamb et al. 2025), with major contributions from enteric fermentation in livestock, manure management, rice cultivation, and residue burning (Smith et al. 2021). Notably, enteric fermentation in ruminants accounts for approximately 90% of global methane emissions from livestock, highlighting its significance as a target for mitigation efforts (Ghassemi Nejad et al. 2024). The urgency of reducing CH4 emissions is emphasized by its potent, albeit shorter-lived, warming effect compared to CO2 (Rial 2024). Therefore, mitigation strategies should prioritize the rapid reduction of emissions from all three major anthropogenic greenhouse gases, including CH4 (Smith et al. 2021).

Quantifying CH4 emissions with precision presents considerable challenges, particularly when addressing diffuse and variable sources such as municipal wastewater systems, natural water bodies like rivers and streams, and anthropogenic reservoirs (Chai et al. 2016). Municipal wastewater collection and treatment systems, for instance, are acknowledged as significant contributors to anthropogenic CH4 emissions (Tong et al. 2024). Nonetheless, actual emissions at the plant or regional level can exhibit substantial variability due to site-specific conditions and pronounced temporal fluctuations (Song et al. 2023). Similarly, global methane emissions from rivers and streams are considerable, estimated at 27.9 (16.7–39.7) Tg CH4 per year, a

figure approximately equivalent to emissions from other freshwater systems (Rocher-Ros et al. 2023). This underscores the pivotal role of land-water interactions in regulating CH4 supply (Rocher-Ros et al. 2023). Recent reviews also highlight that groundwater discharge into aquatic environments is an often overlooked but major driver of global methane emissions, further complicating the estimation of regional carbon budgets (Maglaya et al. 2025). Moreover, human-made reservoirs, often constructed for purposes such as hydropower or irrigation, have increasingly been identified as significant sources of greenhouse gases, including CH4. Emissions from these reservoirs can be particularly notable, especially from hypolimnions where anoxic conditions facilitate methane production (Tabassum-Abbasi et al., 2023).

Accurately estimating CH4 emissions is essential for crafting effective mitigation strategies and upholding the integrity of systems like carbon credit programs ((Ghassemi Nejad et al. 2024). However, traditional methods for estimating methane emissions often face limitations. For instance, in the livestock sector, while direct measurement techniques such as respiration chambers are considered the gold standard, they pose challenges for practical, large-scale farm application (Ghassemi Nejad et al. 2024). In wastewater systems, tier-based the Intergovernmental Panel on Climate Change (IPCC) methodologies may underestimate actual CH4 emissions by as much as two orders of magnitude, especially when sewer emissions are overlooked or when generalized emission factors are used (Song et al. 2023). The lack of detailed information on the precise contributions from various sources, particularly regarding reservoir CH4 emissions, makes it tremendously challenging to develop statistical or analytical models for forecasting such emissions (Ross et al. 2024).

In addressing these challenges, advanced modeling techniques, especially those leveraging artificial intelligence (AI), offer a promising path forward. Artificial neural networks (ANNs) excel at identifying both linear and non-linear relationships and can discern complex patterns even with limited input data, making them well-suited for forecasting CH4 emissions from intricate systems like reservoirs (Abbasi and Abbasi 2021). Furthermore, recent advancements in machine learning, such as Random Forests and Long Short-Term Memory (LSTM) networks, have demonstrated superior capability in forecasting complex environmental and health-related variables, ranging from global rainfall patterns (Wang et al. 2025) to climate-sensitive epidemic risks (Tuan 2025). Progress in AI technology has opened new avenues for more accurately estimating methane emissions in the cattle industry, with machine learning and deep learning algorithms showing potential in handling complex datasets and uncovering hidden patterns (Neethirajan 2024). These models can integrate a variety of data, including animal-related metrics, feed composition, and environmental conditions, to create more precise and robust estimation models (Aguilar-Lazcano et al. 2023). Developing region-specific estimation methods is crucial for improving the accuracy of methane emission estimations, a necessity highlighted for both livestock and other diffuse sources (Tedeschi et al. 2022).

Somalia, situated in the Horn of Africa, is particularly vulnerable to the effects of climate change, as demonstrated by its frequent droughts, food insecurity, and resource-related conflicts (Mohamed et al. 2022). Long-term spatial econometric analyses of Somalia's rainfall patterns over the last 120 years confirm these fragility rhythms, revealing distinct climatic oscillations that complicate resource management (Alasow et al. 2025). This vulnerability is consistent with broader regional trends observed in East Africa. For instance, drought pattern mapping in the neighboring highlands of Ethiopia reveals similar climatic stresses that transcend national borders (Tela and Tariq 2025), while recent analyses of rainfall extremes across Horn of Africa capitals confirm a significant intensification of precipitation volatility in the region (Dhaqane et al. 2025a). Recent analyses in Tanzania have revealed significant spatiotemporal shifts in temperature extremes and rainfall intensity, highlighting the uneven exposure of the region to climatic risks (Beteri et al. 2025). Similarly, long-term assessments in other arid and semi-arid regions, such as Iran, have documented substantial warming trends and drying patterns over recent decades (Hadei et al. 2025), emphasizing the pressing need for climate studies tailored to specific contexts. The nation's economy is heavily dependent on pastoralism and agriculture, which are significant sources of CH4 emissions, primarily due to its large livestock population (Abubakar et al. 2022). The intersection of climate change impacts and rising CH4 emissions intensifies public health vulnerabilities, including respiratory illnesses associated with deteriorating air quality and climate-sensitive diseases (Sen and Chatterjee 2024). Therefore, addressing CH4 emissions in Somalia is crucial for achieving environmental sustainability, enhancing climate resilience, and improving public health outcomes.

Univariate time series models, including ARIMA, ETS, TBATS, Theta, NNAR, and ARFIMA, have been extensively used for forecasting environmental variables (Perone 2021; Thayyib et al. 2023). ARIMA models capture linear dependencies (Sun et al. 2022), ETS models handle trend and seasonality (Thayyib et al. 2023), TBATS address complex seasonality (Naim et al. 2018), and NNAR models capture non-linear patterns (Naim et al. 2018). ARFIMA models suit long-range dependence (Ran et al. 2025). Hybrid models combine these approaches for more accurate forecasts (Patel and Deb 2022; Wang et al. 2024). This

research examines the modeling of annual methane (CH4) emissions in Somalia from 1960 to 2023 using univariate and hybrid time series models. The study aims to identify optimal forecasting methods for CH4 emissions by evaluating their effectiveness. Given the importance of precise CH4 estimation (Ghassemi Nejad et al. 2024; Song et al. 2023) and success of advanced modeling (Abbasi and Abbasi 2021; Ghassemi Nejad et al. 2024), this research will help Somali policymakers address CH4 mitigation and climate resilience.

1.1 Motivation Statement

This research addresses the need to understand and project CH4 emission trajectories in Somalia, a country vulnerable to climate change and health challenges. With its large livestock sector, CH4 emissions are a key national concern, yet forecasting remains limited. Accurate forecasts can inform interventions in agriculture, waste management, and health planning, contributing to climate action plans and sustainable development goals (SDGs) (McElwee et al. 2020). This study bridges the knowledge gap by applying advanced time-series techniques to provide tools for mitigating CH4 emissions and associated risks.

1.2 Novelty of the Paper

This paper introduces a novel comparative analysis of at least six distinct univariate time series models and nine hybrid model combinations for CH4 emissions forecasting specifically within the Somalian context. While similar studies exist for CO2 or other regions (Egeh et al. 2023). This is among the first comprehensive explorations for CH4 in Somalia, integrating the dual lens of climate impact and public health vulnerability. The application of a broad suite of hybrid models to recent CH4 data (up to 2023) enhances the potential for accurate, policy-relevant forecasts.

1.3 Rationale behind the Selection of the Models

ARIMA, ETS, TBATS, ARFIMA, NNAR, and Theta models are selected based on their proven capabilities in handling diverse time series characteristics in environmental data, including trends, seasonality, non-linearity, and long memory (Elseidi 2023). Hybrid models leverage complementary strengths of individual models to capture both linear and complex non-linear patterns in CH4 emissions data (Pence et al. 2024). This selection enables robust comparison to identify the most suitable forecasting approach for Somalia's CH4 emissions.

1.4 Main Objective of the Paper

The primary objective is to apply a minimum of 15 time series models (six single, nine hybrid) to forecast CH4 emissions in Somalia. Model performance will be evaluated using MASE, MAPE, SMAPE, and Theil's U statistics to identify the most effective forecasting model for CH4 emissions in this climate-affected and public health-vulnerable nation.

1.5 What Will this Paper achieve?

This study evaluated six univariate and nine hybrid time series models for forecasting CH4 emissions in Somalia. The objective was to identify suitable models for predicting Somalia's CH4 emissions through rigorous assessment. The study provides forecasts of CH4 emissions, offering insights into future trends and implications for climate policy and public health. It highlights connections between CH4 emissions, agricultural practices, climate vulnerability, and public health in Somalia. The findings offer guidance for policymakers in developing targeted interventions for CH4 mitigation and adaptation strategies.

2 Materials and methods

2.1 Study Design and Data Source

A longitudinal time series design was employed to examine annual CH4 emissions data for Somalia from 1960 to 2023, comprising 64 observations. The data was sourced from the reputable international database Our World in Data (OWID), which offers country-level estimates of methane emissions. CH4 emissions are typically reported in kilotons (kt) or metric tons. The dataset was partitioned into a training set (1960–2013) and a testing/validation set (2014–2023).

2.2 Data Preprocessing

Data were examined for missing values and outliers. Considering the potential for non-stationarity in long-term emissions data (Liu et al. 2022), stationarity was evaluated prior to modeling.

2.3 Stationarity Testing with ADF and PP-tests

In analyzing time series data, stationarity is crucial, meaning properties remain constant over time. The Augmented Dickey-Fuller (ADF) test and Phillips-Perron (PP) test evaluate stationarity. ADF test uses autoregressive model to examine unit root presence. The null hypothesis indicates

non-stationarity through unit root, while the alternative suggests stationarity. ADF test statistics are compared to critical values. If statistics are more negative than critical value, the null hypothesis is rejected, showing stationarity. The ADF test statistics are calculated as follows:

$$ADF = \frac{Coefficient\ Estimate - 1}{Standard\ Error} \tag{1}$$

The Phillips-Perron (PP) test checks for unit root presence in time series, examining both drift and trend unlike the ADF test. The null hypothesis states the series contains a unit root, indicating non-stationarity, while the alternative suggests stationarity. If test statistics are more negative than critical value, the null hypothesis is rejected, showing stationarity. The computation of the PP test statistics is as follows:

$$PP = \frac{Coefficient\ Estimate - 1}{Standard\ Error} \tag{2}$$

The ADF and PP tests produce p-values reflecting the probability of observing the test statistic under the null hypothesis. When p-values fall below a significance level (0.05), the null hypothesis is rejected, indicating stationarity. These tests are widely used to assess data stationarity and guide modeling decisions.

2.4 Software and Research Process

The R statistical software (version 4.4.0) with packages like forecast (Hyndman and Athanasopoulos 2018). tseries, TSA, xts, plotly, astsa, forecastHybrid, and TsDyn were utilized. The process involved: (i) data preparation and time series formatting, (ii) stationarity assessment and transformation, (iii) model development and selection using AIC, (iv) model performance evaluation, and (v) forecasting CH4 emissions.

2.5 Time Series Model Development

The study employs a range of univariate time series modeling techniques to forecast CH4 emissions in Somalia. These techniques include:

2.6 ARIMA Model

The Autoregressive Integrated Moving Average (ARIMA) model is widely used for forecasting time series data, integrating autoregressive (AR), differencing (I), and moving average (MA) components. This model captures both autoregressive and moving average patterns within a time series. The ARIMA (p, d, q) model is represented by the following formula:

$$(1 - \varnothing_i L - \varnothing_2 L^2 - \ldots - \varnothing_P L^P)(1 - L)^d y_t = \left(1 + \theta_i L + \theta_2 L^2 + \ldots + \theta_q L^q\right) \epsilon_t \tag{3}$$

where (y_t) is the time series, (L) is the lag operator, ($\varnothing_i$) and (θ_i) are the autoregressive and moving average coefficients, (p) is the order of the autoregressive component, (d) is the degree of differencing, and (q) is the order of the moving average component.

The ARIMA models were developed using auto.arima() from the "forecast" package to determine (p, d, q) parameters. The model was chosen based on the Akaike Information Criterion (AIC) for model fit. The auto.arima function used the Hyndman-Khandarkar algorithm for selection. As documented (Hyndman and Athanasopoulos 2018), this algorithm involves steps in model selection. The final model had the lowest AIC. Model assumptions were evaluated using standardized residual plot, autocorrelation function plot of residuals, and p-values from the Ljung-Box test.

2.7 ETS Model

The ETS model predicts time series data by incorporating error, trend, and seasonality components in additive or multiplicative combinations. Within the state-space framework, ETS integrates error (E), trend (T), and seasonal (S) components, offering 30 combinations to examine time-series data and address heterogeneity. The E component may be additive (A) or multiplicative (M), while trend (T) and seasonal (S) components can be A, M, or absent (N). The trend component can also be dampened additively (Ad) or multiplicatively (Md) (Sun 2020). The state-space equations for the ETS model are represented as follows(De Livera et al. 2011):

$$Y_t = w\left(X_t - 1\right) + r\left(X_t - 1\right)\epsilon_t \tag{4}$$

And

$$X_t = f\left(X_t - 1\right) + \left(X_t - 1\right)\epsilon_t \tag{5}$$

In these equations, the coefficients w, f, and g denote the respective parameters, while $_t$ represents the Gaussian white noise series. Equation (4) demonstrates the relationship between the lagged state variable $X_t - 1$ and the observed variable Y_t, whereas Eq. (5) delineates the transitional equation that describes the evolution of states over time. To implement the ETS model, the ets() function from the "forecast" package was utilized (Sun 2020).

2.8 TBATS

The TBATS model (Trigonometric Seasonal, Box-Cox Transformation, ARMA Errors, Trend, and Seasonality) is a highly adaptable tool for forecasting time series data. It handles seasonal patterns, trends, and error structures through trigonometric functions for multiple seasonal cycles, Box-Cox transformation for non-constant variance, and ARMA errors for residuals. The formula for the TBATS model depends on its configuration (Karabiber and Xydis 2019). In this study, the TBATS models were represented by the following equation (Perone 2021; Shaub 2020).

$$Y_t^{(\omega)} = L_t - 1 + \varnothing b_t - 1 + \sum\nolimits_{t-1}^{T} S_{t-Mi}^{(i)} + d_t \tag{6}$$

In this equation, $Y_t^{(\omega)}$ represents the observation after applying the Box-Cox transformation with parameter ω, while Y_t is the original observation at time t. L_t denotes the local level, φ represents the damped trend, b represents the long-run trend, T indicates the seasonal pattern, $S_{ti}^{(i)}$ denotes the i-th seasonal component, Mi represents the sea sonal period, and d_t corresponds to the ARMA(p, q) process for the residuals. The TBATS model was identified using the tbats() function included in the "forecast" package.

2.9 Theta Model

The Theta model, also known as the random walk with drift model, is a simple yet powerful method for time series forecasting. Future values are determined by a linear function of the most recent observed value, enhanced by a drift term. The formula for the Theta model is:

$$Y_{t+1} = (1 + \theta)\, y_t + \mu + {}_{t+1} \tag{7}$$

Where $(y_t)$is the time series, (θ) is the parameter representing the slope, (μ) is the drift term, and $({}_{t+1})$ is the error term.

2.10 NNAR

The NNAR (Neural Network Auto Regressive) model is a forecasting tool that uses neural networks to identify patterns in time series data. By combining autoregressive components with neural networks' non-linear capabilities, the model improves forecasting accuracy. The NNAR model uses lagged time series values as inputs into a neural network. This model consists of three layers interconnected by acyclic linkages. The equation for the NNAR, as presented by (Yusof and Kane 2012), is:

$$Y_t = \omega_0 + \sum_{j=1}^{Q} \omega_j \cdot g\left(\sum_{i=0}^{P} \omega_{i,j} y_{t-i}\right) \tag{8}$$

In the equation, Y_t denotes the output, while $y_{t-i}, \ldots, y_{t-p}$ represents the input. The model parameters are represented as $\omega_{i,j}$ (for $i = 0,1,2,\ldots,P$ and $j = 1,2,\ldots,Q$) and ω_j (for $j = 0,1,2,\ldots,Q$), which are referred to as connection weights. The number of input nodes is denoted as P, while the number of hidden nodes is indicated by Q. For seasonal data, the NNAR model can be expressed as $NNAR(p,P,K)_m$ where p represents the previous observed values from the same season used as inputs, P represents lagged inputs, K represents the number of neurons (nodes) in the hidden layer, and m represents the number of months. The "forecast" package provides the function nnetar, which was utilized to fit the $NNAR(p,P,K)_m$ model. In the case of seasonal time series, the default value for P is set to 1, and the value of p is chosen based on the optimal linear model fitted to the seasonally adjusted data. The nnetar function (Hyndman and Athanasopoulos 2018) was employed with the default setting, where K was set to $\frac{p+P+1}{2}$..

2.11 ARFIMA

ARFIMA (Autoregressive Fractionally Integrated Moving Average) model: ARFIMA extends ARIMA by incorporating fractional differencing, capturing long-range dependence in time series. This model effectively analyzes time series data with long-term dependencies. The formula for the ARFIMA (p, d, q) model is:

where (y_t) is the time series, (L) is the lag operator, $(\varnothing_1)$ and (θ_i) are the autoregressive and moving average coefficients, (p) is the order of the autoregressive component, (d) is the degree of fractional differencing, and (q) is the order of the moving average component.

2.12 Hybrid Modelling Approaches

This study examines individual time series models and hybrid approaches that combine techniques to improve forecasting accuracy. Hybrid models were developed by integrating techniques like ARIMA, ETS, TBATS, Theta, ARFIMA, and NNAR. These models, including ARIMA-ETS, ARIMA-TBATS, ARIMA-Theta, ARIMA-NNAR, ARIMA-ETS-Theta, ARIMA-ETS-NNAR, ARIMA-Theta-NNAR, ARIMA-ETS-TBATS, and ARIMA-TBATS-NNAR, were constructed using functions in the "forecast Hybrid" package in R. This approach leverages different models' strengths to capture linear and non-linear trends and seasonality. These functions identify the best-fitting

model. The detailed process of model development using these functions has been previously described (Gunasekera et al. 2017; Panja et al. 2023).

2.13 Model Evaluation

To evaluate time series models for Methane (CH4) emissions in Somalia, the dataset was split into training and validation sets. The training set comprises initial data, with testing containing the most recent 10 observations. Six univariate time series methods—ARIMA, ETS, TBATS, Theta, NNAR, and ARFIMA—were identified and checked against statistical assumptions. Each model produced forecasts for the validation set. Accuracy was assessed by comparing forecasts with observed CH4 emissions using Mean Absolute Percentage Error (MAPE), Mean Absolute Scaled Error (MASE), and Symmetric Mean Absolute Percentage Error (SMAPE). These metrics provide standardized measures of forecast error, with lower values indicating better performance. The best fitting time series models will be determined by evaluation results, focusing on lower error metrics. This evaluation serves as a baseline for developing hybrid forecasting models. While trend analysis would be part of broader interpretation, this step focuses on predictive accuracy for CH4 emissions forecasts. Identifying accurate forecasting models for CH4 emissions is essential for generating reliable projections that inform policy and environmental strategies in Somalia. The evaluation ensures chosen tools are robust and suited to the data characteristics. This study evaluates time series models through data splitting, model fitting, and accuracy metrics to identify effective techniques for CH4 emissions in Somalia, before exploring hybrid models.

2.14 Forecast and Model Performance

To evaluate model predictive performance from complete and training datasets, we utilized error metrics including MASE, MAPE, SMAPE, and Theil's U statistics (Hyndman and Athanasopoulos 2018; Panja et al. 2023). MAPE could not be calculated due to its sensitivity to time-series data scale. Scaled errors such as MASE were recommended as alternatives to MAPE, being unaffected by data scale (Hyndman and Athanasopoulos 2018). MASE is considered a reliable measure of forecast accuracy, suitable for comparing individual and combined time series (Panja et al. 2023; Perone 2021).

3 Results Data Description

3.1 Descriptive Statistics of the Data

The descriptive statistics offer a comprehensive overview of the CH4 emissions data for Somalia spanning the years 1960 to 2023. The variable "X1" denotes the CH4 emissions. The dataset comprises 63 observations. The descriptive statistics are as follows:

The descriptive statistics in Table 1 summarize Somalia's Methane (CH4) emissions across 63 observations. CH4 emissions average 19,258,388 metric tons, with a standard deviation of 4,407,074 metric tons. The median emission value is 19,902,678 metric tons, higher than the mean, indicating negative skew. The trimmed mean of 19,507,170 metric tons lies between the mean and median, suggesting slight skewness. The Median Absolute Deviation (MAD) is 5,009,479 metric tons, indicating variability less susceptible to outliers. CH4 emissions range from 10,332,279 to 25,692,824 metric tons. The data show slight negative skewness (−0.42), and kurtosis of −1.01 indicates a platykurtic distribution. The standard error of the mean is 550,884.2 metric tons, estimating the sample mean's precision. These statistics reveal the central tendency, variability, and distribution of CH4 emissions.

3.2 Visual Representation of the Data

Figure 1 illustrates the line graph representing the trajectory of Somalia's CH4 emissions between 1960 and 2023. The data depicted in Fig. 2 clearly indicate a consistent and uninterrupted rise in Somalia's CH4 emissions from 1960 to 2023.

3.3 Data Stationarity Tests

We employed Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests to assess CH4 emissions dataset's stationarity before model fitting. Stationarity is vital for time series forecasting, ensuring statistical properties remain constant over time. Results in Table 2showed the CH4 time series achieved stationarity after one differencing technique, confirmed by both tests. This ensured our modeling of CH4 emissions in Somalia was conducted on stationary data.

Table 2 presents stationarity test results using ADF and PP tests. The ADF test yields a p-value of 0.1801 for original data, exceeding 0.05 significance, indicating non-stationary

Table 1 Descriptive statistics of CH_4 emissions for Somalia (1960–2023)

Variable	% Obs. (*n*)	Mean	SD	Median	Trimmed Mean	MAD
Data	63	19,258,388	4,407,074	19,902,678	19,507,170	5,009,479
Variable	Minimum	Maximum	Range	Skewness	kurtosis	SE
Data	10,332,279	25,692,824	15,360,545	−0.42	−1.01	550,884.2

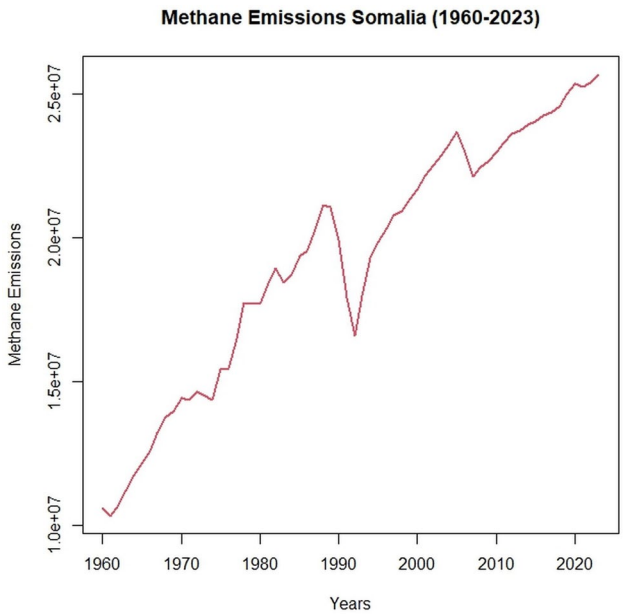

Fig. 1 Methane (CH_4) emissions in Somalia, 1960–2023

Table 2 Stationarity tests for the CH_4 emissions series: augmented Dickey–Fuller (ADF) and Phillips–Perron (PP)

Test	Test Statistics	*p*-value
ADF (original data	−2.9745	0.1801
ADF (first differencing data)	−5.3054	0.01
PP (original data)	−11.811	0.4119
PP (first differencing data)	−35.385	0.01

CH4 emissions. After first differencing, ADF test shows −5.3054 statistic and 0.01 p-value, suggesting stationarity. PP test for original data shows −11.811 statistic and 0.4119 p-value, confirming non-stationarity. For first differencing data, PP test shows −35.385 statistic and 0.01 p-value. Based on both tests, first-differenced CH4 emissions series is stationary, while original series is non-stationary, indicating one order of differencing is necessary for modeling.

3.4 Timeseries Model fitting, Evaluation of Model performance, and Forecasting

After differencing the CH4 emissions dataset for stationarity, six single and nine hybrid models were tested. Model specifications like ARIMA (identified as ARIMA(0,1,1) with drift from Fig. 2) were selected by minimizing AIC

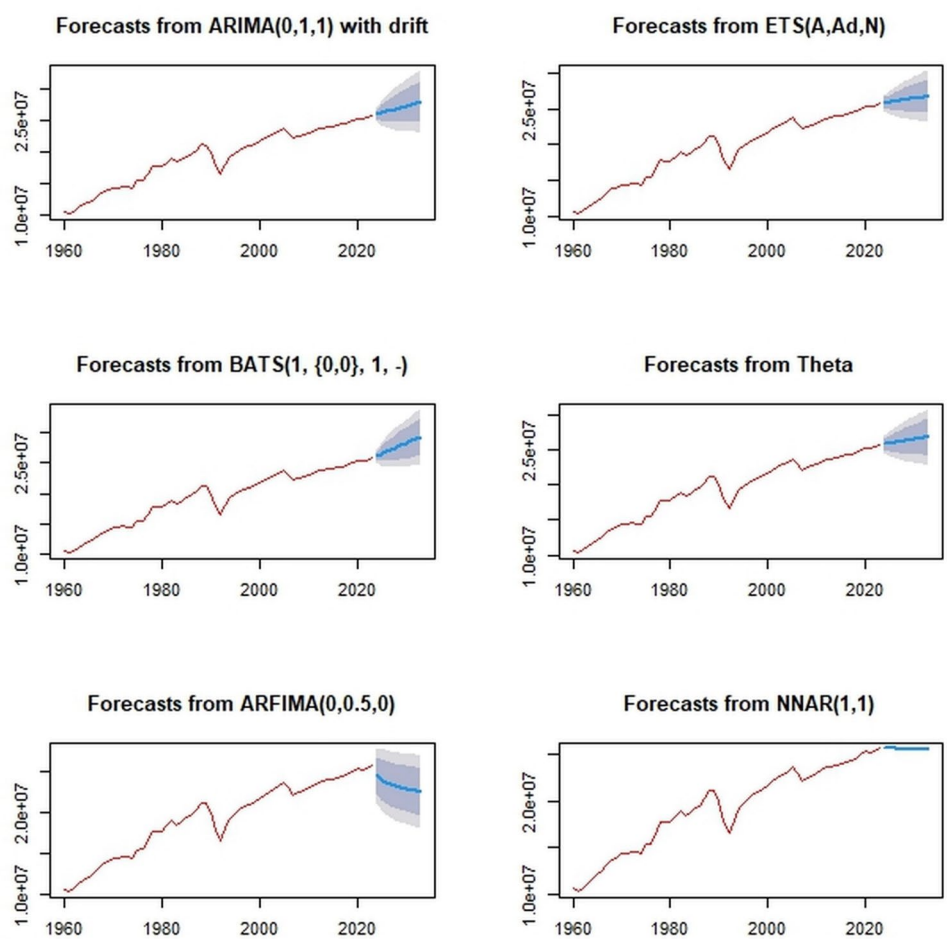

Fig. 2 Forecasts of CH_4 emissions from single models (ARIMA, ETS, TBATS, Theta, ARFIMA, NNAR); blue line indicates model forecasts

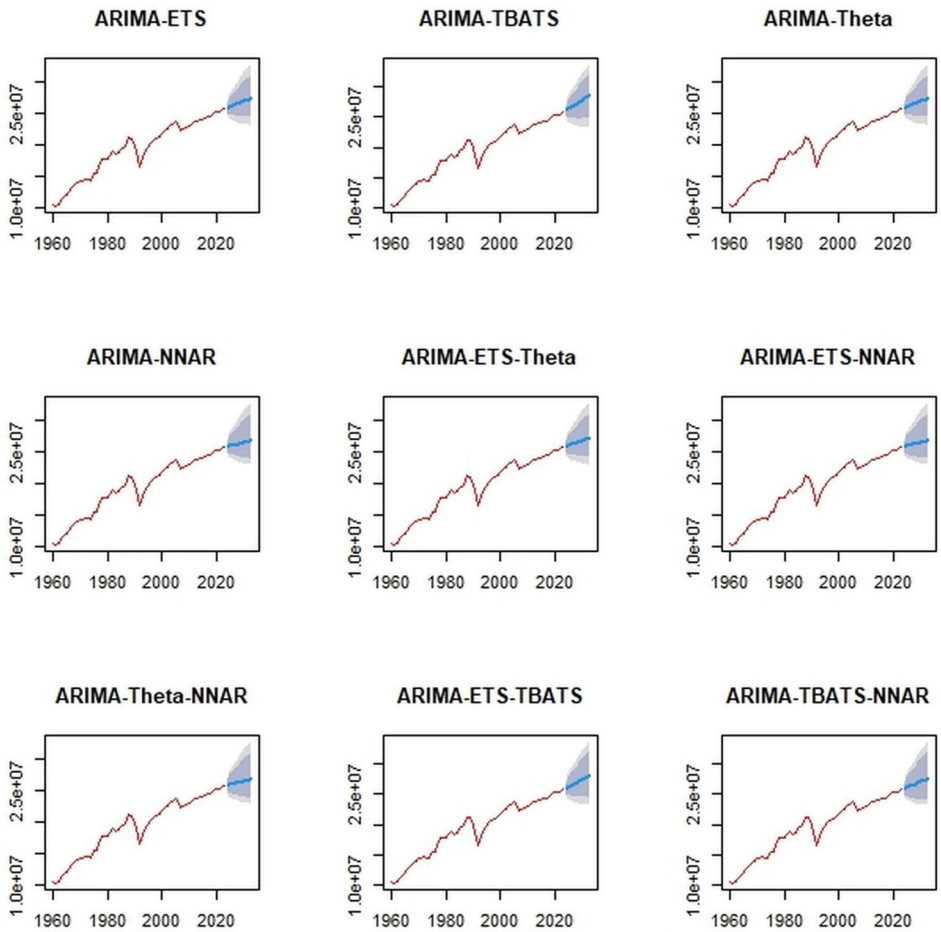

Fig. 3 Forecasts of CH_4 emissions from hybrid models (ARIMA–ETS, ARIMA–TBATS, ARIMA–Theta, ARIMA–NNAR, ARIMA–ETS–Theta, ARIMA–ETS–NNAR, ARIMA–Theta–NNAR, ARIMA–ETS–TBATS, ARIMA–TBATS–NNAR); blue line indicates model forecasts

and checking residuals for violations. For other models including ETS (ETS(A, Ad, N)), BATS (BATS(1,{0,0},1,-)), Theta, ARFIMA (ARFIMA(0,0.5,0)), and NNAR (NNAR(1,1)) (specifications from Fig. 2), appropriate fitting procedures were used. Figure 2 shows actual, fitted, and forecasted CH4 emissions values, with year on x-axis and emissions in metric tons on y-axis. The forecasts predict an upward trend in Somalia's CH4 emissions.

Figure 2 showcases the corresponding actual, fitted, and forecasted CH4 emission values for the nine hybrid models developed in this study. These hybrid approaches combine techniques like ARIMA with ETS, ARIMA with TBATS, ARIMA with Theta, ARIMA with Neural Network Autoregression, and three-way combinations like ARIMA-ETS-Theta or ARIMA-TBATS-NNAR (from Fig. 3). As with single models, the x-axis denotes the year, and the y-axis represents CH4 emissions. The plots demonstrate each hybrid configuration's performance in tracking historical data and projecting future trends. Most hybrid models predict a notable upward trajectory in CH4 emissions over the forecasted period.

Table 3 presents the model performance values for the single time series models, including the Theil's U, Mean Absolute Percentage Error (MAPE), Symmetric Mean Absolute Percentage Error (SMAPE), and Mean Absolute Scaled Error (MASE) statistics. These metrics serve as indicators of the models' forecasting accuracy. In contrast, Table 4 Forecast accuracy for hybrid models (ARIMA–ETS, ARIMA–TBATS, ARIMA–Theta, ARIMA–NNAR, ARIMA–ETS–Theta, ARIMA–ETS–NNAR, ARIMA–Theta–NNAR, ARIMA–ETS–TBATS, ARIMA–TBATS–NNAR) on the 2014–2023 test set; metrics reported are Theil's U, MAPE, SMAPE, and MASE; lower values indicate better performance; best results in bold displays the corresponding performance values for the hybrid models.

Table 3 Forecast accuracy for single models (ARIMA, ETS, TBATS, Theta, NNAR, ARFIMA) on the 2014–2023 test set; metrics reported are theil's U, MAPE, SMAPE, and MASE; lower values indicate better performance; best results in bold

Model	Theil's U	MAPE	SMAPE	MASE
ARIMA	**0.8664**	**0.6596**	**0.0065**	**0.3165**
ETS	2.7891	2.2969	0.02265	1.1019
TBATS	3.4104	2.7072	0.0266	1.3020
Theta	1.9729	1.4799	0.0149	0.7145
NNAR	6.5309	5.2751	0.0545	2.5382
ARFIMA	6.4125	5.1004	0.0527	2.4563

Table 4 Forecast accuracy for hybrid models (ARIMA–ETS, ARIMA–TBATS, ARIMA–Theta, ARIMA–NNAR, ARIMA–ETS–Theta, ARIMA–ETS–NNAR, ARIMA–Theta–NNAR, ARIMA–ETS–TBATS, ARIMA–TBATS–NNAR) on the 2014–2023 test set; metrics reported are theil's U, MAPE, SMAPE, and MASE; lower values indicate better performance; best results in bold

Model	Theil's U	MAPE	SMAPE	MASE
ARIMA-ETS	1.8018	1.4254	0.0141	0.6843
ARIMA-TBATS,	2.1083	1.6439	0.0163	0.7905
ARIMA-Theta	**0.7471**	**0.4835**	**0.0049**	**0.2334**
ARIMA-NNAR	2.9469	2.3806	0.0241	1.1453
ARIMA-ETS-Theta	2.3329	1.8526	0.0183	0.8902
ARIMA-ETS-NNAR	1.1239	0.8189	0.0082	0.3951
ARIMA-Theta-NNAR	2.5986	2.0626	0.0209	0.9932
ARIMA-ETS-TBATS	2.3329	1.8526	0.0183	0.8902
ARIMA- TBATS-NNAR	0.9354	0.6851	0.0069	0.3297

Table 5 Forecasts of CH_4 emissions for 2024–2033 from the best single model (ARIMA(0,1,1)+drift); columns show point forecast and 95% prediction interval (lower, upper)

Model	Year	Point Forecast	95% lower bound	95% upper bound
ARIMA	2024	25,902,618	24,945,798	27,995,630
ARIMA	2025	26,135,722	24,275,814	28,818,929
ARIMA	2026	26,368,825	23,918,721	29,525,413
ARIMA	2027	26,601,929	23,678,444	30,165,274
ARIMA	2028	26,835,032	23,504,790	30,760,596
ARIMA	2029	27,068,136	23,375,675	31,323,430
ARIMA	2030	27,301,239	23,279,048	31,861,209
ARIMA	2031	27,534,343	23,207,476	32,378,902
ARIMA	2032	27,767,446	23,155,990	32,880,025
ARIMA	2033	28,000,550	23,121,075	27,995,630

The tables provide a comparative analysis of the models' performance using these different evaluation metrics.

The results in Table 3 show that the ARIMA (0,1,1) with drift model outperformed other single-technique methods across all forecasting metrics. It achieved the lowest Theil's U (0.8664), MAPE (0.6596), SMAPE (0.0065), and MASE (0.3165), providing the most accurate forecasts for CH4 emissions in Somalia. Models like NNAR (MASE = 2.5382) and ARFIMA (MASE = 2.4563) showed higher error values across metrics, indicating poorer predictive performance.

Table 4 compares nine hybrid time series models for Somalia's CH4 emissions forecasting. The ARIMA-Theta hybrid model showed superior accuracy with lowest error metrics: Theil's U (0.7471), MAPE (0.4835), SMAPE (0.0049), and MASE (0.2334). These results indicate the smallest forecast errors in magnitude and percentage deviation. Other hybrids, like ARIMA with Neural Network Autoregression, showed larger errors (MASE of 1.1453). Based on Table 4; Figs. 1 and 2, the ARIMA-Theta hybrid model outperformed both single and hybrid models across all metrics, surpassing the best standalone ARIMA (0,1,1) with drift model (MASE = 0.3165). This demonstrates the advantage of hybrid approaches combining different forecasting techniques. The ARIMA-Theta hybrid model is recommended for CH4 emission forecasts in Somalia. Figures 1 and 2, and Tables 3 and 4 reveal projected CH4 emissions trajectories, with both superior models indicating an upward trend in emissions over the forecasted period, suggesting concerning implications for Somalia's future methane footprint. Tables 5 and 6 present the forecasted CH4 emissions from the best single model (ARIMA (0,1,1) with drift) and best hybrid model (ARIMA-Theta) for 2024–2033.

Table 6 Forecasts of CH_4 emissions for 2024–2033 from the best hybrid model (ARIMA–Theta); columns show point forecast and 95% prediction interval (lower, upper)

Model	Year	Point Forecast	95% lower bound	95% upper bound
ARIMA-Theta	2024	25,854,972	24,542,899	27,071,754
ARIMA-Theta	2025	26,028,784	24,133,765	27,995,630
ARIMA-Theta	2026	26,202,595	23,846,459	28,818,929
ARIMA-Theta	2027	26,376,407	23,622,219	29,525,413
ARIMA-Theta	2028	26,550,218	23,438,284	30,165,274
ARIMA-Theta	2029	26,724,030	23,282,980	30,760,596
ARIMA-Theta	2030	26,897,841	23,149,369	31,323,430
ARIMA-Theta	2031	27,071,653	23,032,934	31,861,209
ARIMA-Theta	2032	27,245,464	22,930,537	32,378,902
ARIMA-Theta	2033	27,419,276	22,839,890	32,880,025

Table 5 shows that the ARIMA model projects CH4 emissions to increase from approximately 25.9 million metric tons in 2024 to around 28.0 million metric tons by 2033.

Similarly, Table 6 indicates that the ARIMA-Theta model forecasts CH4 emissions to rise from approximately 25.85 million metric tons in 2024 to about 27.42 million metric tons in 2033. Both models project a steady increase, with the ARIMA model showing a slightly higher endpoint forecast for 2033. These quantitative forecasts, along with their 95% confidence intervals, provide crucial data points for anticipating future emission levels and underscore the importance of proactive mitigation strategies.

4 Discussion

This study aimed to evaluate and compare various time series forecasting models encompassing both single and hybrid methodologies to project annual methane (CH4) emissions in Somalia. As a nation deeply affected by climate change and confronting significant public health vulnerabilities, understanding the trajectory of these emissions is critical. Our analysis utilized CH4 emissions data spanning from 1960 to 2023, divided into training (1960–2013) and validation (2014–2023) sets, to project emissions for the decade 2024–2033. This focus is particularly relevant given that agriculture, specifically livestock production, constitutes

the cornerstone of Somalia's economy and serves as the primary anthropogenic source of CH4 (Abubakar et al. 2022; Smith et al. 2021). Accurate quantification and forecasting of these emissions remain essential yet challenging tasks due to the complex nature of environmental data (Ghassemi Nejad et al. 2024; Ross et al. 2024).

The initial phase of our analysis prioritized data integrity by evaluating stationarity. Using Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests, we identified the original CH4 series as non-stationary—a common characteristic of long-term environmental datasets (Abbasi and Abbasi 2021). To address this, we implemented first-order differencing to achieve stationarity. This rigorous approach to establishing robust trend baselines is critical in climate-sensitive studies and aligns with recent climatological assessments in other arid and semi-arid regions; for instance (Hadei et al. 2025) employed similar rigorous statistical trend analyses to decode decadal climate variability across Iran. Achieving stationarity is a prerequisite for reliable time series modeling, ensuring that the subsequent forecasts are statistically valid (Hyndman and Athanasopoulos 2018).

In the modeling phase, six single time series models (ARIMA, ETS, TBATS, Theta, ARFIMA, and NNAR) were compared against nine hybrid combinations. Among the single models, the ARIMA (0,1,1) with drift emerged as the superior standalone approach, demonstrating an ability to capture linear dependencies effectively (Zhang et al. 2022) However, environmental data often contains non-linear complexities that single models may miss. In our evaluation of hybrid models, which leverage the complementary strengths of different modeling paradigms (Patel and Deb 2022; Wang et al. 2024), the integration of ARIMA with the Theta method (ARIMA-Theta) proved to be the most accurate. This hybrid model achieved the lowest error values across all metrics (Theil's U, MAPE, SMAPE, and MASE), surpassing all other configurations.

The superiority of the hybrid approach in this study underscores the benefits of combining linear and non-linear modeling techniques to capture varying trends and seasonal dynamics (Pence et al. 2024). This finding is strongly corroborated by recent global research demonstrating the efficacy of advanced computational methods in environmental forecasting. The utility of combining traditional statistical approaches with artificial intelligence models has proven particularly effective for resource assessment in semi-arid regions, as demonstrated by recent applications in North African aquifer systems (Bouri et al. 2025). For example, Wang et al. (2025) utilized machine learning approaches to significantly outperform traditional linear models in forecasting complex global rainfall patterns, while Tuan (2025) highlighted the superior predictive capability of AI-driven models over standard statistical baselines in predicting climate-sensitive health risks. Our results suggest that hybrid models like ARIMA-Theta offer a similar leap in accuracy for national-level emission forecasting.

The projection of CH4 emissions in Somalia for 2024–2033 using these optimal models offers crucial insights for policymakers. The ARIMA-Theta model predicts a consistent rise in emissions, reaching approximately 27.42 million metric tons by 2033. These projections, illustrated in Figs. 1 and 2, highlight a concerning upward trend. These challenges are not unique to Somalia but reflect a broader pattern across the Horn of Africa and East Africa. Recent analyses by Beteri et al. (2025) in Tanzania indicate similar shifts in climatological trends, where rising temperature extremes and rainfall variability pose significant threats to agricultural stability and economic development. For Somalia, a nation dependent on pastoralism, these rising emissions signal a need for urgent intervention. Anticipating these trajectories allows for targeted mitigation strategies in livestock management (e.g., feed quality improvement) and waste management. Furthermore, accurate forecasting is essential for determining regional contributions to the global carbon budget and establishing reliable background levels, a necessity emphasized by Malacaria et al. (2025) in their modeling of atmospheric methane and carbon dioxide in the Mediterranean basin. By integrating these forecasts into national planning, Somalia can better align its climate adaptation plans with National Determined Contributions (NDCs) and the Sustainable Development Goals (SDGs). Just as recent hydrological modeling in the Juba River Basin has highlighted the necessity of scenario-based planning for water allocation (Dhaqane et al. 2025b), our findings underscore that robust time-series forecasting is equally vital for managing national methane emissions.

It is important to acknowledge certain limitations. Time series forecasts rely on the assumption that historical patterns will continue; however, external factors not modeled here—such as abrupt policy changes, severe climatic shocks (droughts/floods), or technological breakthroughs in methane mitigation—could alter future trajectories. Additionally, while data was sourced from reputable international databases, uncertainties in national-level inventories for developing countries remain a challenge (Saunois et al. 2020). Future research should aim to integrate multivariate variables, such as specific livestock population dynamics, land-use changes, and socio-economic indicators, to further refine these models. Despite these limitations, this study identifies the ARIMA-Theta hybrid model as a robust tool for decision-making, providing a vital evidence base for enhancing climate resilience in Somalia.

5 Conclusions

We performed a comparative study of time series models to predict annual methane emissions in Somalia, a country facing climate and health challenges. Our evaluation included six individual time series models (ARIMA, ETS, TBATS, Theta, ARFIMA, and NNAR) and nine hybrid model combinations, assessed using forecasting accuracy metrics. Among the single time series models, the ARIMA model (specifically ARIMA(0,1,1) with drift) showed the strongest performance. When considering all fifteen models, a hybrid model combining ARIMA and Theta methods (ARIMA-Theta) demonstrated the highest forecasting accuracy, surpassing all other models tested. This highlights the effectiveness of hybrid strategies in capturing the complex dynamics of environmental emissions data. The successful prediction of CH4 emissions in Somalia for 2024–2033 using optimized models offers vital insights for policymakers and officials. Projections from both the top single model (ARIMA) and the hybrid model (ARIMA-Theta) suggest a continued increase in Somalia's CH4 emissions. This trend presents significant challenges to environmental sustainability and public health, given the agricultural sector's contribution to CH4 production and the country's climate vulnerability. The findings enhance understanding of CH4 emissions forecasting in regions with limited data and provide evidence for informed decision-making to reduce emissions in Somalia.

5.1 Recommendations

To tackle the issue of increasing CH4 emissions in Somalia, the study suggests several recommendations: Policymakers in Somalia should focus on creating comprehensive national strategies to cut down CH4 emissions, especially in key areas like livestock farming and waste management. Efforts should be directed towards sustainable livestock practices that reduce methane emissions, such as improved feeding techniques, manure management for biogas production, and climate-adaptive pastoralism. It is crucial to boost national capabilities for monitoring, reporting, and verifying (MRV) greenhouse gas emissions, including methane (CH4). Enhanced data quality will lead to more accurate forecasts and better policy development. CH4 emission predictions from the ARIMA-Theta model should be integrated into national development plans, climate change adaptation strategies (NAPs), and international commitments (NDCs). Given the connection between CH4 and environmental degradation as well as public health concerns, it is vital to strengthen surveillance systems to track climate-sensitive diseases and respiratory conditions related to poor air quality. Encourage the adoption of renewable energy to reduce biomass burning and implement waste-to-energy technologies to lower methane (CH4) emissions from landfills.

Finally, promote international partnerships to share practices, expertise, and resources for CH4 mitigation, especially for cross-border issues like pastoral mobility. Support research into CH4 emissions by incorporating socio-economic factors into forecasting models, examining emission hotspots, and assessing mitigation strategies. Update models with new data to improve forecast precision.

5.2 Future Directions

Additional research is needed to enhance CH4 emission prediction models in Somalia. Improvements should include multivariate models integrating livestock data, land-use changes, climate information, and socio-economic factors. Sector-specific emission models focusing on enteric fermentation, manure management, and waste sectors would enable precise policy measures. Spatial analysis could identify emission hotspots for region-specific strategies. Policy effects and scenario-based forecasting could assess mitigation impacts. Machine learning algorithms and model ensembles show potential for improved predictions. System dynamics modeling could examine relationships between emissions, agriculture, climate, and health outcomes. Enhanced emission inventories and regional comparisons with Horn of Africa countries can promote collaborative mitigation efforts, supporting Somalia's sustainable development initiatives.

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Author Contributions All authors contributed to the study conception and design. Conceptualization, methodology, software, data curation, and analysis were performed by Mohamed Abdi Abdullahi. The first draft of the manuscript was written by Mohamed Abdi Abdullahi. Supervision and critical revision of the manuscript were provided by Dr. Abdisalan Hassan Muse, Hamse Abdi Aden, and Saralees Nadarajah. All authors commented on previous versions of the manuscript. All authors read and approved of the final manuscript.

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Data Availability The datasets analyzed during the current study are publicly available from Our World in Data; processed data underlying the findings and analysis files are available from the corresponding author on reasonable request. While direct validation of climate data with ground-based observations for Somalia within the scope of this study is challenging due to data limitations, the 'Our World in Data' sources are widely recognized and used in climate research.

Declarations

Competing Interests The authors have no relevant financial or non-financial interests to disclose.

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